

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIP506D - ADVANCED WATER RESOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The porosity of a sedimentary rock primarily depends on:
A. Degree of metamorphism
B. Sorting and grain packing
C. Magnetic susceptibility
D. Chemical weathering
- 2 The validity of Darcy's law is mainly limited by:
A. Fluid temperature
B. Laminar flow conditions
C. Aquifer depth
D. Well diameter
- 3 The primary function of a well screen is to:
A. Increase pump efficiency
B. Prevent fine particles from entering the well
C. Improve water chemistry
D. Reduce well depth
- 4 In rotary percussion drilling, the bit action mainly involves:
A. Grinding only
B. Impact and rotation
C. Suction
D. Chemical dissolution
- 5 A properly completed well ensures:
A. Minimum casing
B. Maximum drawdown
C. Structural stability and controlled inflow
D. Elimination of drilling fluid
- 6 Artificial recharge is most effective when the aquifer is:
A. Highly cemented
B. Very shallow
C. Unconfined with high permeability
D. Located below clay layers

- 7 Groundwater quality deterioration is least influenced by:
 - A. Agricultural return flows
 - B. Saltwater intrusion
 - C. Pump efficiency
 - D. Industrial effluents

- 8 The need for artificial recharge arises mainly due to:
 - A. Increased rainfall
 - B. Declining groundwater levels
 - C. High porosity in all aquifers
 - D. Excess canal water

- 9 Groundwater occurrence in igneous rocks is usually controlled by:
 - A. Bedding planes only
 - B. Fractures and joints
 - C. Presence of fossils
 - D. Organic content

- 10 Percolation tanks are typically constructed in:
 - A. Steep hill slopes
 - B. Low-permeability basalt
 - C. Hard rock terrains with shallow weathered zones
 - D. Deep confined aquifers only

- 11 In mathematical groundwater models, boundary conditions are required to:
 - A. Reduce aquifer thickness
 - B. Define flow behaviour at model edges
 - C. Increase computational cost
 - D. Eliminate storage terms

- 12 Which hydrogeological region of India is known for extensive hard rock aquifers?
 - A. Indo-Gangetic plains
 - B. Peninsular plateau
 - C. Assam valley
 - D. Coastal alluvium

- Q2. Solve the following.
 - A) Define porosity and permeability of rocks and discuss the factors affecting porosity. 6
 - B) Describe how the electrical resistivity method is used in groundwater exploration and state its major advantages. 6

- Q3. Solve the following.
 - A) Explain the use of type curves in pumping-test analysis and mention their limitations. 6
 - B) Define Darcy's law and describe its key components as used in well hydraulics. 6

- Q4. Solve Any Two of the following.
 - A) Describe the steps involved in well development and explain its importance in groundwater extraction. 6
 - B) Define rotary drilling and rotary percussion drilling, highlighting differences between them. 6
 - C) Explain the importance of regular maintenance of wells and discuss the consequences of poor maintenance. 6

Q5. Solve Any Two of the following.

- A) Define conjunctive use and explain its significance in addressing water-resource challenges. 6
- B) Discuss the role of uncertainty analysis in regional groundwater modelling and how uncertainty can be quantified. 6
- C) Explain the major factors influencing groundwater quality and outline practical measures for improving it. 6

Q6. Solve Any Two of the following.

- A) Describe how check dams and percolation ponds contribute to groundwater recharge. 6
- B) Describe various artificial recharge methods used in groundwater development. 6
- C) Discuss the challenges associated with groundwater depletion and how integrated recharge methods help mitigate them. 6

***** End *****